ActInf GuestStream #026.1 ~ "From Users to (Sense)Makers" and Stigmergic Annotation

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all right hello everyone it's august 12 2022 and we are in guest stream number 26.1 from users to sensemakers on the pivotal role of stigma social annotation in the quest for collective sense making this should be a great presentation and discussion and so i'll pass it to ronin to introduce the context and take it away with the presentation so thanks everyone for joining and ronan and will thanks for coming on awesome yeah thanks daniel for for putting this on and uh yeah excited to be here and have this uh discussion i think it's our first time discussing this paper like long form so looking forward to it um but yeah this paper actually code yeah for some context this presentation will be related to the paper that we wrote together as a group um and yeah it was also represented in the conference on hypertext and social media in june of this year in barcelona uh so today we'll go over kind of the key points of the paper and uh try to dive into more details uh kind of explore directions that we didn't have time to address with this sort of short kind of presentation of the paper of the conference um and then we'll later sort of open a wider discussion uh talk about how this is sort of informing the work of our collective and discussion of some initial projects that we're working on in the context of this of this research including active inference and some other projects like vo and a few more so yeah looking forward to a lot to talk about so yeah probably we should start with the definitions or some of the terms we're going to be using in this because we introduced a lot of a lot of words that can mean a lot of things to a lot of people so i think yeah sense making is something that has a lot of different missions the one worst of working working definition we're looking at processes by which individuals and groups make sense of their environment by organizing and send data to the environment is understood well enough to enable reasonable decisions so we do science making all the time in physical environments and online environments as we'll see so there's this kind of incoming stream of sense data coming from the environment and we have to sort of uh adapt and handle this this kind of stream of data to inform our decisions uh incorporate it into our decision making process um but yeah so scent making is is a kind of broad subject in terms of like where where collectively descent making some kind of uh [Music] a kind of typical sort of uh thing you might hear from you know media you'll hear things like like this what we brought here so yeah the quality of our democratic life depends the public having the fact that being able to make sense of them uh and there's this feeling that this isn't really happening anymore so that's why a lot of people are worried about the future of democracy this is just kind of an interesting uh research here on the right that's showing how polarized and how like how much we don't have a shared sense uh what's going so you can look at things like uh you know the topics people are looking at and seeing on cnn or fox news uh and see like how different the coverage is of each of those uh networks for for the same topic so yeah so things like trump's failures will get a lot of uh a lot of uh being addressed like very heavily in cnn but uh not seeing much on fox and then vice versa democrats so yeah people are living in a really different world because of our information environments so this is kind of a motivating background for why we're why are we getting interested in this because really our democracies are at stake as a lot of people have already noticed and yeah just kind of another word of sort of disclaiming that stem making is a wicked problem um and yeah we definitely aren't going to claim to have like the solution to making our collective sense making it will require many different solutions and it's kind of an ongoing problem so we'll kind of make progress on it and then we'll have new problems progress it's very interdisciplinary so there's a lot of stakeholders of different sort of schools of thought that and practice that will need to be involved so we're looking at kind of we're trying to look at one angle that is how platforms shape our information landscapes online and and we kind of noticed two things two uh sort of important things about platforms that make them what we call it perilous so for one there's a notion of centralization where a few giant platforms facebook people youtube etc have gained unprecedented control over attention by controlling the needs to create search and distributed information and two opacities where platforms network data and algorithms are hidden away from the public so there's like all the sort of oversight that could help us you know issues with our information landscape or really not we don't really have uh we don't really have any way to to handle this currently because of this opacity yeah the kind of dual scrolling has become the icon of our times and we don't know what's behind our screens like why isn't something being shown to us we kind of have to at times you know vague or like the different motivations for why that might be but yeah we really are kind of yeah they're controlling your screens and we're on those screens for most of the day so definitely a lot of power over what we're our information types so you know it's clear how we got here right it's kind of convenient you know we have these platforms that we have tik tok and twitter they offer us they do offer us like very high quality in some regards like they give us information that we're interested in we like seeing you know what we're looking at on instagram like seeing the videos we're watching on kickback they give us like this short-term dopamine kick but of course then when we look into the global picture we can start seeing the cost of this the platforms are involved in a host of deeply problematic social phenomena like polarization information epistemic extraction and yeah it's interesting to sort of go into the literature scientific literature on this because uh you know scientists yeah this is just just one uh uh tweet that i am very particular you know the most important paper of my career talking about this kind of dramatic restructuring of the human communication not by these platforms with no aim other than selling ads uh yeah the paper is called stewardship of global collective behavior it's a very inspiring call to action this is a crisis businessman just like ecology and sustainability or crisis disciplines we need to understand how we're going to help save the planet from ruining nature so we have a similar kind of crisis that's what they're sort of claiming on the front of uh yeah on the front of like the information online online environments uh that people are looking at yeah so there's a lot of work on this so this is you know the scientific side this book is another kind of interesting uh very inspiring book uh that also kind of puts the fingers squarely on this kind of challenging attention economy you know that says that if we're going to surround the minimal challenges we face today uh we'll need to be giving the right field of attention right things the book is called stand out over light talks about freedom and resistance for the attention and these are all very inspiring sort of uh works and the question is where do we go from here so we're really inspired and we understand that the information environments are very sort of putting us in this parallel what can we do so this is also from the same book uh we're kind of working toward where we can start to actually be doing work on helping rethink these information so our quest kind of begins and with this kind of realization and again this is from the book we presented earlier if we're at all serious about promoting freedom and autonomy in the digital age what that would entail starting to assert and defend our freedom of attention this is also a very evocative kind of uh passage freedom of attention yeah what does that look like on online spaces there's a lot of ways you can think about freedom of attention yeah so a lot of this work is is sort of i think attempt to sort of articulate one vision of what freedom of attention looks like uh online uh yeah so you know we have this quest we we we're trying to sort of put this then more into more time people it's what it means as a spoiler you know what we talked about in this in in this presentation what we talked about in the paper uh we look at we take signature to be some kind of unifying concept that's underlying attention and scientific so if we need some place to start sort of a concrete solid place to put our feet and start working uh we think stigma energy is a really useful concept many of you probably haven't even heard of stigmatizing i surely haven't until i met daniel here i learned about you know collective intelligence and insects so yeah we'll start from ground zero stigma g is collective behavioral behavioral coordination among entities via perception and modification of shared environment and this i think really did originated in studies of uh coordination in ants so you know the classic sort of example is trail pheromones and sleeves and we'll go into more detail on that so you know an ant leaves some kind of environment modification in the form of a trail pheromone and then it's attracted to it and there's some this collective intelligent behavior emerging as a result another important thing to say about sigmar g is is this on the next slide yeah so it's really kind of profound when you go into it because you know we shape the environment and it shapes us in a very profound way um some even go as far to say that environments act as a distributed memory system for a collective organism so if any of you are familiar with the extended mind hypothesis in cognitive science how our tools are sort of extensions of our cognition so stigmaty is a sort of extended mind for collectives and then we have these diverging feedback loops where you can go back here you know we have agents and they're producing some kind of mark that's where stigma in greek you know mark is coming from so agents are producing these results of action and that's stimulating you know the same agent or other agent and uh so there's a kind of coupling between the environment the agent acting on the environment stimulating other agents for action uh and then this kind of emerging global sense yeah also important no large scale coordination is possible without the energy so if you go like above n equals 25 then then you need emerging communication it's not like you know we can do zoom calls with a thousand people right we need other methods of coordination synergies there's this like main method of doing it and internet modifying environments that other people are participating in we'll go through a lot of examples but the point is that we really need to create healthy stigmargic infrastructure for a large scale coordination and then yeah we can ask is stigmatic that is provided to us today in online environments how healthy is it uh so we'll talk about that in a minute yeah if you have anything you wanted to say about uh like other aspects of synergy that we may or may not be touching here maybe later on the uh types of stigma yes cool yeah so that's our next slide yeah um yeah we took a lot of integration from this work uh i'm leslie marsh on uh stigmatic epistemology and and they give this kind of useful distinction between two kinds of stigma g and you have content creating stigma g they call it semitectonic i think that's they're drawn from some other uh literature i think oh cool yeah acts as a sign of book because i wasn't sure that actually nice so yeah content creating energy versus signaling signature which is marker based so the examples are helpful here to understand what h mean in a physical environment uh content creating is like actually carving out a trail which other agents are going to see and you know use that uh they take out a different kind of sign and you should go over this trail and not just like randomly off into the woods and in a digital environment you might think of things like writing a document so you're actually modifying the digital environment by writing words so that's kind of the content creating sigmarching and on the other side you have signaling stigma gene the marker base where looking at things like pheromone trails and physical environments so ads aren't actually modifying the environment in any sort of meaningful way i mean they're just doing it as sort of communication experience they're not changing the environment by like moving some debris out of the path or creating a path just like marking some kind of uh trail for other ants in the digital environment you can think of a sort of highlighting or tagging is you know the counterpart to that in the physical environment so we'll definitely go into this also in detail now we are focusing here on this big building based stigma g we kind of highlight that this is what we want to focus on um because actually yeah this doesn't get as much attention thinking about online environments as we'll see a lot of people are talking about this kind of content creating stigma g and sort of people don't mention it but people talk about how we create content how we share it online uh not as many people are talking about it's kind of how we're sharing markers online so this is why we decided uh this is kind of an important focus not going to bring to attention yeah and did you want to say anything else about texas in the physical case the left side the actions that modify the niche and clear out the trail or move pebbles and do the nest architecture and so on the analogy to content creation is very clear like there's some arrangement of mass that isn't where it was previously and then on the right side where in the physical case it's more like the pheromone trails which are transient or longer lasting accumulations and uh as i hope we can draw out on the next slide and going forward there is this kind of accumulation across both types like somebody might make an initial post on a social media but then somebody else might annotate it it blurs the line between what is a reaction what's a pheromone and what is like building another piece of content like if somebody responds with a image is that like building or is that like annotating and marking and i think that points to the need to have a a design language and a way of framing this question that is able to encompass like the pure classical content delivery like a paper publication and pure classical transient marking on some digital context and everything in between so that we don't have to categorize different communication acts along the lines that they've been defined by essentially the platforms that got us into this position yeah yeah yeah no that's that's that's really great and i think here this maybe creates a misleading impression there's some kind of uh pattern categorization it's like taxonomy but yeah there's really a spectrum between marker base and content creating uh so it's kind of a yeah it's there's all the grayers in between and some things some markers are almost content in their own right yeah if i add a highlight with some words on it is that really content creating or is that annotation or like signaling so yeah definitely a lot of a lot of spectrum here and this is just to say that we're focused more on this side of the spectrum and then really we can talk about this spectrum uh give a better idea of what we're looking at so we're looking at the merging markers on the web um and the reason we're focused on them is because in a lot of respects they function as sort of the digital traces of human attention um and you're looking at things like reactions associations that people might make you know hyperlinks like oh i think this content relates to smaller content we do it all the time in our tools like notion you're you know you're linking to some other page you think this reminds me yeah tags some tagging content saying oh this is the on the topic of i don't know this decentralized distance or something like that uh and click through data is interesting this is an implicitly yeah an explicit and implicit marker so some of these markers people are kind of doing this they're you know leaving consciously the digital pheromone trail by liking some posts and other times it's actually like you can think of it as an implicit marker where it's just like the trail that we're traveling on the web uh we're leaving this marker and it is definitely being harvested right like click through data is very valuable as we can we'll see also later the are selling and importing um but yeah there's the difference in like what people people might not think they're even creating any kind of data here people today are more sort of aware of it but uh yeah i think there's a lot of interesting these are some interesting future directions if we're talking about like implicit versus explicit so there might be things that are more implicit you know in the future for example pupil measurements or gaze tracking brain computer interfaces so this is just you know where you're looking on a screen could also be a magic marker given that someone is recording it you know it's getting recorded in some kind of medium and being used to sort of guide someone else's attention yes then that is a stigmatic marker if no one is recording it then it's you could say yeah it's not actually a suggestion there are definitely yeah there's a few things to say about this one is that the reason these markers are so important is because they're very cheap to create but they have a very high information content so this is why you know machine learning um research researchers and practitioners will pay a lot of money for human annotated data annotations like very it can be you know a few bits of information what label is some category uh but it's very high information content because the human looked at this picture and decided okay this is a dog so these things are very high information content and that makes them very valuable and they're also cheap to create and interpret so yeah this makes markers very uh valuable but it also creates kind of challenges with uh ethics user consent uh how the data is being used privacy because this thing is very valuable but on the other hand we're also creating a lot of it without even thinking so should this be ours uh or should they help you guys so a lot of questions there cool so we're getting closer now to certain understanding where our questions going so yeah the emerging workers could be harnessed for healthy collective sense making but centralized platforms are regarding them you know now we kind of can see that if we look now everywhere we look at what platforms are doing we can see that they're actually these dragons that are hoarding worship workers like everything we're doing is getting really recorded without us having much access to it so the click-through data likes all this kind of stuff is is very precious and it's getting collected by these platforms so okay we kind of know where we're going now a little better i want to say one thing this is just a little um yeah go into a little bit of what we mean when we say healthier collective symptoms like why do why do we think that symmetric markers could be harnessed for health or collective sentiment so let's try to give a little more uh insight on like what is healthier unhealthier uh you know information comments so yeah we can look at a few examples just like a toy example of ants i think maybe can be illustrative this is kind of the mental model i had in my mind uh as i was thinking about this paper so you know if you're looking at healthy thicker green ants so an ad you know find some food source um and they start bringing back the nest they lay the pheromone trail that line there sorry for the poor graphics and other ants are going to see that trail be attracted to it and follow it the food sources consumes and meanwhile the the pheromone trail is decaying as the food source is consumed um eventually you know just the case and then you know it's going to go there they need to waste the time because there's no more food there that's kind of how the healthy sort of pheromone for pheromone decay the the ground you know holds it for some time and then there's you know the sort of predictable uh operating characteristics of it now imagine a different kind of a different kind of setup where you know you have a different kind of ground right like here i'm using platform just to be suggestive but you have a platform which is actually like swallowing pheromone okay so now ants are trying to lay the pheromone but it's kind of decay right so this is already one way in which the stigmargic infrastructure is not healthy because uh in a sense like the answer can find a food because it's just not gonna be able they're not gonna be able to lay that parallel there uh to attract other ants another thing that could happen is you know if platforms are over amplifying so if this kind of ground instead of uh decaying predictably it was kind of for some reason reacting and over amplifying pheromones so even when the food source was consumed ants would still be kind of traversing and wasting their time uh even when there's no food there so another thing you know you can kind of see the parallels you know in light spaces uh yeah the third third thing you might think of is a case where you have kind of areas where you can't lay uh uh you know pheromones you can't leave the feral trail so uh if there's food it's outside the area where you can leave pheromone then ants won't be finding it because you know just the properties of the ground so these are all kinds of ways in which there's just trying to give you to sort yeah convey the intuition that there are it's a very sort of delicately tuned system and there are many ways it can be out of tune and which would you know lead to very disastrous consequences for the sort of organism the collective organisms as a whole yeah yeah well do you want us to think about that uh yeah i was just going to observe that looking back on the concept of sort of information silos that you discussed with uh fox news and cnn um maybe one more aspect of unhealthy sigmoidy is some of these stigmargic markers or trails are visible to one group and invisible to another group so you you also have um this failure to create a shared reality um i wonder if that's another aspect of unhealthy big energy yeah definitely i think that one yeah that would definitely be another another a great example where yeah you might have you know that only a certain part of the hive is going to be sensing and then another type that the other part of the hype is going to be something they're really going to be just living in shared shared uh not shared realities yeah for sure so i think that's a that's a great example also yeah those are the few kind of hand maybe intuitive examples try to get the uh point across that if we're looking at healthier information environments uh they're going to be highly complex because if you consider a lot of parameters here you know just imagine what what's going on this is ants so it's relatively simple and even that when it was complex but um you have all these different kinds of markers people are leaving and the different parameters to reach the markers and how they're getting spread used to recommend content so a lot of parameters here the point is that as designers and participants in these information environments uh will need to be able to assume more control more decentralized right because we don't want a central designer but uh we'll need to be able to play with these parameters for uh currently as we discussed there's the opacity and centralization so very few people um are able even to make the kind of you know to tweak these parameters and they can do experiments you know like global scale and they're the only people that can do them so yeah that was the point of these examples so our proposals so finally we can kind of talk about where where we think uh fruitful direction would be to go um yeah we call it open source retention and we try to draw the parallel to open source software and decentralization movement and the sort of as i was alluding to earlier they're all about freeing most of the work and about creating content creation sharing so you have things like github and you know decentralized currencies like bitcoin um all those are examples of you know the technology the content creating stigmatization so we can create code we can create documents we can you know publish social media posts uh all that kind of thing that's content we're talking about open source tension we're looking at sort of the complement of that it's the emerging markers is anyone looking at this idea of freeing reactions to count so we're doing a lot of this kind of work on decentralizing contemplation but we're actually what we're kind of the moment is like we're not actually doing much work on freeing these kind of reactions to count so on the technical level we think that just as we have sort of systems and that will help us with protocols that share our content we that's on the technical level like at a high level and then we have here sort of just like the social aspect of the open source software movement with these kind of hacker heroes that were building code in public and uh sharing it without sort of any monetary compensation uh then we'll have a similar kind of uh we envision kind of a similar kind of scent maker hero they're sort of the counterpart and they're and they're doing kind of curation in public and they're mindful of their digital attention traces so doing kind of open source attention open source thinking sort of following these lines of open source coding um yeah so let's look at what open source attention means for like uh information ecosystems like how we translate that to the information ecosystem actually we're zooming in now we're getting into the sort of more in the direction like concrete uh implementation so first let's understand like the platform-centric information because there's all the new terms we have the new terminology so what we can see like okay so these are the dragons that are hurting our attention data and how it happens is we have these kind of now we can sort of see what's happening here the the do controlling we're doing here is kind of a low agency annotation interface uh that's collecting our submerging markers uh platforms are storing this data they sell it to advertisers they use it to drive content discovery in a way that serves optimizing for platform groups by serving us new content that will keep us on our screens as well as ads that will keep the advertisers happy so they can sort of this is the feedback right they have this they they have the the control closed on us because they have the data and they have the algorithms how might you like do something alternative that doesn't involve those dragon platforms uh so what we propose is a decentralized maker centric ecosystem so instead of users we have makers and makers are these kind of high agency annotators so we can think of all these personal knowledge management tools uh notion and rome and you know we'll look at vio hopefully later next all these different kinds of annotation interfaces people are building these knowledge management tools you can actually think of them as annotation interfaces that are collecting emerging markers uh but the crucial sort of point is that we're talking about a case where makers are owning these markers this is all this is this is you know the point of sort of stealth sovereign storage uh cell power and data uh you have things like solid protocol that allow people to actually control and own their own data and we're saying if we apply this testing markers we actually get this separation uh decoupling of the content discovery algorithms and the mystic merchant markers then you kind of break up this platform loop and then content discovery services will need to be sort of uh focused and they'll need to be aligned with take into account the sort of human-centered uh incentives so makers will be able to share data with these content discovery services and there'll be a lot of diverse patent assembly services but they won't have control over our data and then that's kind of a way to break up that vicious cycle so you can imagine you know different kinds of feed algorithms that you'll have like a competing market free algorithms or you'll have matchmaking services uh ai recommendation all kinds of um diverse new discovery services and and you can use all of them or any of them and you know kind of it incentivizes open source yeah open source and more human-centered environment than what we're looking at previously one of the main challenges this is like one of the first challenges that proposal is running into is how to scale up because you know we have this kind of network effect where we need to compete essentially with facebook and twitter and all these huge established networks because they have the data to you know provide the high quality recommendation and we don't have anything right now so the idea here is to bootstrap growth and personal knowledge management tools for collective knowledge management this is an idea that's kind of been floating around in a lot of different spaces but yeah this kind of feeling that user curator knowledge is really stuck in single-player tools and and all the while search engines really aren't doing as well as they could be doing how do we build more collectively curated knowledge that works not just like these silos second grade so um i think one way to look at it is that from going sort of this is swing depending on the other direction where if previously there's data silos in the platform levels so each platform is like a data style using that for harmful sort of purposes uh this kind of problem is here what's currently happening is like each of these small relatively small knowledge management tools is going to be a data style and they won't be large enough to sort of reach yeah critical map they will be large enough to provide high quality content discovery so they have some data but it's not enough so all of them are competing versus each other um either none of them are going to be able to provide like this kind of uh high enough quality service and they won't be able to beat the platforms the big platforms so what we kind of proposed here is is a protocols not platforms uh approach and this is something that's very sort of widely advocated for web three so you see protocols for all kinds of all kinds of sort of interoperable objects and what we're talking about here are protocols for uh merging primitives so you might imagine a protocol for light or backlink or other kinds of things we talked about earlier uh and that what this effectively would enable is having markers uh being able to you could share markers essentially so instead of having these dialogue markers you can actually pool them so if i'm collecting my reading list in notions or also doing it in rome and someone else is doing it in another app they could also be shared in a common format and use for driving this kind of discovery that is kind of where we're envisioning it going we took a lot of inspiration from this uh very good uh essay on protocol stock platforms highly recommended so yeah i think kind of the meme firm meme version of that is we have these kind of current situations where these small independent platforms are are going to be eaten by the big fish to doc and twitter and facebook and all those and the way for these independent platforms to coordinate is through interoperability using protocols for stigmatic markers that's kind of our vision but yeah there's a lot of work to do to get there um so some next steps we are thinking about are definitely kind of going deeper into the weeds and a lot of questions remaining about social incentives uh the tech stack uh malicious use of these kind of technologies uh cultural transitions like how do we do the transition from users to makers work that the first this kind of paper was more like a blue guy idea paper and it was more like functioning as a call to action um to the sort of challenges we're saying we need to start building tech for governing our attention which is sort of something that we haven't heard much talked about so we wanted to sort of bring that uh emphasize that in this paper uh yeah we are really looking for all kinds of collaborators so check out our website and connect with us if any of us interested um i think we can yeah maybe pause there open for discussion awesome questions thanks for the presentation ronan so for those who are watching live feel free to write a comment in the live chat otherwise we'll if you have any reflections or thoughts and then we have a few more slides to go through in this next steps phase and that'll include looking a little bit more at some platform affordances and also talking about how this connects to active inference no immediate thoughts here um plenty to talk about when we get to the discussion and the questions let's go to uh the uh promising directions and talk about that and then we'll go after that to vo and then active inference yeah great um so yeah some promising directions we kind of became aware of this is actually even after after we wrote the paper um this is an interesting word from a few years back on opportunities and challenges around public web activity tracking so essentially getting people to look at experiments human subjects and explored the kind of affordances for allowing people to share browsing activity traces for example like urls i visited and time spent on each webpage information like that so they gave people a tool that would allow them to do that inside the browser and they checked like small small groups of people um different kind of friend groups or research groups and they kind of lot of interesting things i think uh some of the highlights were that they found that like while people had privacy concerns they definitely can and will tailor their sharing depending on the concept context and people felt that this made them a lot more mindful of their browsing behavior in a lot of times in a good way which kind of speaks to the user-to-maker transition uh if people are sort of being open so it's just like you write open source code you kind of make sure it's like high quality because we know people are looking at it so they may lose like attention and browsing and people also could kind of feel how this would help them better sort of navigate the information landscape because they were saying like you can better assess the legitimacy of some website by knowing how many people visited it and who visited this kind of uh shared you know public information um yeah this was this book kind of caught my eye uh yeah if you ask me i would rather a thousand a billion times that researchers for an academic and humanistic knowledge purposes are the ones that can map information stored for humanity rather than private companies so this is really speaking kind of people feeling that sort of human traded church humans or humans can be healthier information departments definitely this work also pointed to sort of the limitations you know it's like how do we scale up to something something larger than just a group of friends so yeah but definitely this is like a cool intro in preliminary study um yeah vo uh we have also definitely a very interesting kind of preliminary work that will tell us more about yeah thanks ronan um vo is it's a prototype stage software project um that was actually originally designed to be a platform that is a kind of knowledge garden um a way for people to visually build their network of references and make connections and trace their paths back um and it was in a sense a way of creating trails for people between uh pieces of content in different platforms or silos and of course the problem with that as uh as ronan has discussed is it itself is in a cycle and is in a platform so it's a little bit self-defeating it's in its original conceptualization um so right now we're actually in a stage where we're sort of going back to the drawing board taking this this new research this new approach of building a tool that works on an interoperable protocol using distributed data instead of a centralized uh data store with proprietary mechanisms of interaction and hopefully we can take vo and make it a a test case where we can practically apply the concept of this common sense protocol to build out a piece of software where the data can then be used in other pieces of software other front ends other search engines and so on and in that way start to actually break free of this um platforms dominate our attention paradigm and see kind of what the possibilities and limitations are we have a short demo that is kind of the original concept of vo um so we'll go through that real quick i don't think the sound will play but just full screen the video and just play it and will describe us um yeah okay i got it yeah yeah if you want to describe well you can kind of say what we're seeing here all right so this is an example of how vo might work so it takes you through a process discovery you can essentially keep track of what you're interested in learning um go through the entire search process look for sources that seem interesting or relevant to you and then as you go you can go through and annotate different pieces um take detailed notes take excerpts etc and then as you have new questions as there are new uh areas that you would like to search you can essentially capture those as you go without breaking your uh train of thought without breaking your research process and then from there vo automatically builds a visual map of what you've explored and what you have yet to explore so that you can continue to build an ongoing map and then when you come back to this many months later you have the entire context of your whole research process visually laid out for you everything you looked for everything you found everything you were interested in but never got around to so you kind of take a lot of the burden of manually managing your recent process off of the researcher and allow the technology to do what the technology does best um so largely the way for you to take kind of a walk at your own face through the information you're interested in or is valuable to you at the level that you can understand it so it allows you to sort of self-select the best learning path um without really getting lost in platforms and algorithms and i guess content suggestions or even curated content or courses that might not be at the proper level of depth for where you are at the moment um go ahead yeah oh did you ask the thing well go first run and then i'll add a few notes um i just wanted to say like this this is um i mean that was one of the things that sort of i think triggered the whole collaboration and like got us started on this on this direction of research um like seeing what vo can like you know just playing a few intriguing things like one is that many annotation tools don't provide uh this kind of temporal relation between annotations so they're just like you're annotating everything and you don't really know the structure of like how you got there and where you went from there and things like laid out in the trail forum was really evocative and and sort of thinking about like how would this look like we could share this kind of information and like imagining the sort of multiplayer video where you might be standing on a node and then you're like seeing the trails of other people and maybe it's like even using ai to sort of highlight the people that are maybe you know recommended that you should follow their trails based on stigma g and crowd signals and ai or whatever um so like that was very like a yeah one of the thought there about yeah that you know just yeah i think there's another thing i want to say like social vio and the sort of linking annotations or like two seemed like really really intriguing directions to take this yeah i also um saw that transition between the first person knowledge management and the second person that ronan had brought up and it hinges on shared landmarks and so being able to think of these in a spatial context which maybe we can return to with active inference and then being able to go down to just one's own path like just like we'd have a trace on google maps or something like that but then see multiple paths together and have curation approaches there and then i think it's really uh it's a fascinating tension with this functionality the single or the multiplayer case it could be done in a centralized platform web 2 strategy and there's even arguments for why that would be streamlined more efficient more secure all these different aspects yet in a different way or at a higher level there are challenges and failure modes and so on that would be like dismantling the function potentially for example allowing for a centralized point of control over those path recommendation algorithms so as long as the architecture is bundled then there are certain failure modes that this is actually like escalating to the next level people say oh we used to have the news feed it was linear and now we have the branching news feed and it's so much more fun and engaging but then it's just representing the next escalation in attention capture yeah that's a really good observation oh yeah go ahead uh yeah that's that's kind of why we're we're motivated to pull out of this platform paradigm um and kind of follow this uh this new approach it's fundamentally like all all we're trying to do is enhance people's ability to make sense of the world and if we make a tool that undermines it no matter how cool or fun it is we failed um bringing it into a context where this data is you can take the same data and use it in different apps or you can take data from different apps and use it in here and you can kind of break free of certain mechanisms where the fact that the data is siloed could be used against the end user and instead create a paradigm where this information can be acted on and interpreted in a lot of different ways where there's room for innovation and insight and creativity and competition and disruption where this model can be evolved outside of just this app by one team of developers and can be integrated into different ways of interacting with information um by freeing it from that then then the concept isn't only serving a business or a group a small group of people it's serving everybody who stands to benefit from the concept being available um so that's kind of what we're trying to do we don't want to shoot ourselves in the foot and in the collective violate our purpose yeah yeah yeah yeah i was just thinking i just thought i was thinking of like like you were talking about the different kinds of apps like i mean by letting this beta be free it opens up like the space of like you don't really know how to use i mean you know it's incredibly valuable you can't even envision like how it could be used in different ways i'm just thinking of for example you know like learning learning uh apps i think there's like a lot of intriguing space there to explore like you know the trail that someone is making in the beginning of their steps and some topic is very different to the trail that someone was making like an advanced trail and you could if you could learn these different kinds of trails you could build like really cool online curriculum um services that's one thing i was thinking of but there's just like so many others uh and i think there's just a lot of questions you know people are maybe thinking themselves like yeah i mean obviously there's technical questions because what we're saying here just to kind of drive the point home we're saying we don't think veto should be the only tool that will let you map these rabbit holes we want a lot of different tools to be able to do this and interoperate with each other that's one thing that's kind of okay because it raises a lot of confusing questions about you know okay how do you guys monetize this stuff like where are where you know where is the data getting stored and how people not feeling your data or yeah how are you maybe getting uh compensated for people using your data in different kinds of services um so we're aware that there are a lot challenging questions there like we're not naive about that but we do think that these kind of questions are joining you know broader questions in web three space of data sovereignty and decentralized identity and yeah all the bigger questions that a lot of people are working on and we see ourselves sort of interfacing with them and you know people might in a modular way like if someone has a good solution you know renumerating people on their data stations yep bring it to active inference and maybe go to the second of the active inference slides uh first and break it off awesome let's get back here and the next slide we'll come back to the general questions but i just wanted to start with this one um highlight some awesome recent work and provide like an example based entry point to the intersection of attention which is often used very abstractly or in a single entity case in active inference or in a non-stigmatic setting how attention is being used and bring it to like a very clear applied case so this is the recent work of albarasan at all from this year and they are making a model of kind of like a toy twitter where individuals are involved in just like in any active inference model they're involved in perception cognition and action and then impact in the niche and in this case what is being perceived from the niche what the generative process is handing those entities is information that in this case is being proxied by different hashtags so you know yes on x no on x of course hashtags have much more complex usage simply tagging content or using or not using one can see many ways that the hashtag is not the whole semantic picture but again it serves as a proxy and as an expansible platform to make models that could include different kinds of communication on the left is a a graphical representation of how this simulation plays out and so a simulation like this can be used prospectively given a set of parameters to evaluate different futures it can also be used given empirical data to tune those parameters and in fact there can be kind of a bi-directionality or a tale of two cities tale of two densities where empirical data are coming in and updating parameters and then those updated parameters are used in a generative capacity again to look at counterfactuals or to simulate futures and it's called a graphical model because it's based upon a bayesian graphical network model the nodes whether they're a circle or square there's some details to go into but the nodes are variables so that would be like x equals those are the nodes and that would be equivalent to like a variable in a script so whether you think about this more mathematically or more from a programming an implementation angle the nodes are like the variables and the edges represent relationships amongst the variables and for those who have more familiarity with this type of representation of an active inference model they'll see what some of these letters mean and what roles they might be playing here but suffice to say that this graphical network here is representing the cognitive context of multiple entities that are receiving information in the form of hashtag doing some cognitive process on it updating their beliefs as well as choosing how to act and then implementing that action and then that is like one time step in the simulation so i wanted to provide this example because it's a great case study of bringing active inference into a social and digital setting and it's a great direction that uses also the pi mdp package developed by some of the authors and others so it's a great case on multiple sides to understand how research developments and open source software development can meet up with socially relevant cases so wanted to just describe that case if either of you have any thoughts or questions and then the next case that we'll look into will be more of an ant case and that will help bring us to some of these bigger questions about attention and stigma chi yeah i was just going to say um this kind of circles back i think to the one of the first lines we were talking about is this kind of stewardship of global collective behavior paper which was kind of uh you know pulling out the fact that there's no ethical or scientific oversight over the large platforms and then saying this is something like how you know how how an oversight might look and actually it's more than over there right because actually it's even deeper than oversight because it's actually like you might imagine this being involved in sort of intervention in platforms just being able to uh if you have recommendation services this could like inform uh you know how to update parameters the recommendation service based on observed behavior of the network uh so yeah beyond oversight even sort of like this kind of collective decentralized control we were alluding to so i think this is like a more concrete instance of how that might look a very fascinating kind of direction a modeling direction that people can maybe find interested and it also really highlights the relevance of a bottom-up agentic perspective not just because we then can talk about pluralism and diversity of perspectives much more easily but even the kinds of design interventions and suggestions are going to be different if we think about this from like the top-down content depositing or the bottom-up approach like imagine if we wanted the ants in the colony to be in a different location and we're putting all these articles in one location we're making so much amazing content promoting this idea but of course if the entities are not navigating to it and reinforcing that trail you could spend all the money on advertising and it would just dissipate as soon as that like life support for the trail were gone and so when we take that bottom up attention-based perspective it really puts us like in the view from the inside from the participant what are they perceiving which is always going to be extremely limited and then what are they updating their beliefs in terms of and then how are they choosing to act given the affordances they have so the bottom-up perspective brings in all these very discussions around like accessibility in discourse and perspective integration and then treats it as like something that might be addressed at the level where it's occurring which is at the entity or the actor level rather than being like well there's 500 petabytes of data and this is how much the system does on this high level those are abstract metrics again with a with an economic analogy it'd be like macroeconomics and then behavioral and like neuro and micro economics one can just put piles of product in different locations but unless there's individuals who are interested in it and prefer it and allocate their attention to it nothing will occur maybe another just like to try a concrete example for that could be you know a will brought up this uh then the dashboard of the fox news versus cnn so you might imagine like you know stewards of some futuristic uh social network or you know social i look at the kind of attention data we're talking about collecting here and they might be able to through these kind of modeling approaches because okay you know we what if we want to optimize this network uh for you know rent coverage and like get people to see diverse viewpoints and stuff like that like how are the which recommendation services we use and how should we tune them uh yeah and then sort of seeing the effect of that and be able to learn how to collectively steward attention and you know again this open source people can participate in it yeah in directions that are more healthy for society one last thought before going to the next slide is um in the vo traces we saw about how what was being presented or provided or curated wasn't necessarily the granular content so not the newsletter that says here's the five things you want to look at or hear ranked by whatever metric novelty or by importance or popularity here are the granular atomic pieces of content for you to look at when we can curate and in in a consensual way determine how transitions amongst content are coming into play that's like the true value okay so you're on the wikipedia page for active inference or you're on the wikipedia page for some historical entity maybe that content is just so amazing it stands alone but the reality is it's part of our epistemic forging our learning journeys and so being able to connect dots is quite literally understanding okay for somebody who is in a given context and language experience this intro to python should link to this next resource but then of course there's many links so there's no reason to collapse that pathway to like a hallway it kind of respects the actual dynamics of the information environment in a few interesting ways um let's go to the next slide because it it connects some more points yeah the back ones are backwards or uh the forwards okay so um in 2021 with some colleagues i worked on a ant colony simulation where the individual nest mates were utilizing an active inference model of perception cognition and those actions were spatial movement in a tea maze so in a foraging space and then one other action that they had the opportunity to do upon finding food was to deposit pheromone so no pheromone was being deposited until the food was detected and then upon kind of turning around a pheromone deposition occurred and it was decaying in the environment it's a kind of fun link to look at the place where that core model of epistemic foraging being foraging in a space and looking for something seeking for something having a preference to find something that we used in the ant spatial foraging case for finding the food resource in the teammates it was adapted from a earlier slash different paper by axel constant at all and the paper was acquisition of culturally patterned attention styles under active inference and so the setting is very creative and subtle which is that there's a tic-tac-toe grid so a three by three grid that is representing like the visual scope of someone who's looking at something so s um staring or they're foveating on the central point and then they have like a limited range of peripheral vision and then outside of that there's much less visual acuity they were involved in an epistemic foraging task so unlike where in reward learning the decisions of where to move and how to move might be dictated by what is perceived as the most pragmatically rewarding state or paths towards pragmatic reward in this information foraging case which has been also um explored a lot of other eye movements active inference simulations it's driven by epistemic foraging reducing uncertainty about what kind of pattern one is existing in and so it's almost like reading braille except it's with our eyes because we of course don't see at the same resolution the whole page at the same time in fact the clarity outside of your main visual region and the clarity even in your blind spot is part of our generative model of vision but putting that aside what the entities are doing in this pottery scanning task is actually scanning locally and using the local patterns that they're seeing with a relationship of like okay is it like a line going across or line going up or is there a diagonal understanding the motifs that they're perceiving to reduce their uncertainty about what kind of pottery they're looking at and this the only tweak main tweak that we had to make to bring it into a spatial foraging context was thinking okay instead of the at the center of the visual gaze instead of that just being like the x y coordinate of where one is looking at we're going to make that the body position of the nest mate and so visual foraging you can scan and you can you can like teleport or move very rapidly but it doesn't leave a trace you could look over something many times and it's not going to modify the object in the ink case because it's this embodied spatial foraging they are only able to move locally and they do under all slash sum situations physically modify the niche and i think it really closes the loop with what ronin brought up earlier with like eye gaze and other brain computer interfaces are taking invisible stigmergic markers of attention and making them into visible markers like heat gaze analysis or heat maps of gaze not sure what they call it it is often used like in advertising um where people might present two versions of a page or a product and see where people are looking in terms of the label and other features do we have or use that kind of approach for education do we proactively use it do people field test different layouts of papers so that's an interesting thought and um oh yeah i just want to yeah i definitely feel like uh the whole i was thinking just like um yeah i feel like a whole new sort of yeah field of epistemic engineering for like there's so many both like them you know fury engineering yeah practice they're all at all different levels there's just like yeah it's like pretty wide blue motion space for different kind of techniques modeling this is really cool i like how you bridge between sites like the site where we have here like the affiliation and the ant movement like it's really interesting like how and each hand is sort of like a sensor of the collective organism yeah it almost felt like what we're trying to build and for social networks kind of like we're trying to build like a seismograph or something like that like uh epidemic seismograph yeah by using making this kind of what you're saying the individuals or another kind of visual mixed metaphor if there's a top-down attention determiner it's like the eye of mordor in lord of the rings it's like a laser that's like leaving like a a a scar where it's scanning so it could be modifying the environment or not but it's like one strong laser versus like seeing as like many distributed eyes that are in local feedback loops so cognitive burden if we think about the actual biology of it or their computational burden if we think about simulating this system would be lower and more paralyzable and wouldn't require any entity to have access to information that we know it can't have like how do people interact with markets well they don't have the time series data from the last 10 years loaded into memory but the analyst might but the individual is actually on this journey seeing random things from different times and so if we wanted to understand decision making in that setting we would want to be taking or at least complementing the top down with a bottom-up view and so it's the case in economics and it's also the case in our epistemic environments though those are um corralled in a way that keeps the discourse on certain areas yeah i well we kind of talked about a little before i feel like this there's another whole wrap hole to talk about how how much you want to bring in this sort of machine aided decision making because we understand that like humans like on one side i really i can see where you're going with like the sort of how local agents are making sense of local environments that's sort of creating this global more global trends making uh but then i know we also know that humans like when when we choose for example you know what path from this like social view we're eventually we're going to use ai to sort of integrate histories of people that have traversed those paths besides you know here suggest like a few to go on so i think that's just the delicate uh balance between maintaining human agency and interpretability that like know what they're doing they're not just like following blind visa instructions from algorithms versus really needing algorithms to make sense to crunch the data but doing it in a way it doesn't sort of i think that i've heard it's called like yeah algorithms that disintermediate humans i think it's kind of like the human is like a peripheral part of the sense the sense making good yeah there's definitely like a delicate block to walk there let's go let's go to the first um active inference slide and then so we have kind of two slides of general questions to raise some of them to to bring us out from those more specific examples some general active inference questions and then we can um close with the open challenges so just the first one was what is attention in active inference and how is it being used because there's a technical usage of attention that's common in bayesian statistics like you could hear about a neural network with an attention module or even a simpler kind of model with attention as a parameter and it relates to how much new observations are updating your priors so if you have no attention on an observation it's not updating your priors so that is equivalent to not paying attention it's like there's things you're not seeing behind you you're not paying any attention they're not being used up to your priors whereas if you're super paying attention to something then you're going to be updating your priors to its location or what it's um telling you so where have we seen that kind of motif in active inference and how did it play a role in the alberosen at all paper and how does it play a role in these kinds of second questions like a very general question that we ask all the time which is just where and how is active inference being applied are those conceptual or schematic simulations or are these real-time data processing systems which almost inevitably are going to require technical solutions that aren't like just the active inference entity loop that's kind of like the kernel but then there's a technological system that uses the kernel just like the linear model kernel like y equals mx plus b that is so simple can be written down yet in practice there needs to be a lot of architecture and multilinear aggressions and so on so this is a different kernel for modeling this kind of an active situation and also the systems need to have more than just an insightful kernel to be of any utility let alone function the third point we recently had a symposium on active inference and robotics where we had about uh nine presenters who share different work that is involving the realization of robotic systems implementing active inference and to see the kinds of uh topics that they were bringing to the front like affective robotics robots that are dealing with us as affective entities but also robots that even have computational emulations of affect like what if a robot could have a frustration parameter but it used that to not go down a road that it wasn't really finding value in for example or boredom if the information wasn't good or excitement if it were good so those kinds of affects have a technical and computational usage in active inference models and then i think the the last question that brings us to the more open questions and a lot of the other threads that our co-authors have been bringing in is like where is design in this and humor human-centered design graphical design and the user interface but taking that as like the tip of the iceberg and how do we bring in proactive thinking around what kinds of systems and experiences that are to be curated how does that design thinking meld with cognitive science in a positive way cool yeah i one thing you mentioned i think this was in the beginning when we started working on this research you asked this question it kind of stuck with me you said something about like yeah we have like what are we doing with all the cognitive science theories if we're not using them like like you're saying that effect it's like yeah they're not we're not implementing the theories in practice like often when they're building products it seems yeah this seems like a really interesting kind of theory integrating with practice to build yeah information landscapes uh yeah very very i think cognizant has a lot to say here i'm excited to see how this gets kind of integrated i like how it kind of gets gets a little uh meta too so you have i i like what you said earlier you said epistemic engineering i would i'm i'm kind of inclined to reframe it as like epistemic design epistemic environment design in which agents are using their understanding of active inference to develop systems that allow individuals to interact with the world in a way that's used as active inference so it's sort of like and i mean that's kind of what um i believe that i've heard it phrased as active open-mindedness that's a sort of like self reflection it's this this ability to evaluate your own cognitive processes update your models and your own cognitive processes in ways that allow you to interact with other external information so that you can continue to update your models where they might otherwise become entrenched in overconfidence and confirmation bias and whatnot and this is kind of taking it out of that individualistic paradigm and applying it in a collective paradigm where you're you're by designing uh ecosystems epistemic ecosystems that allow people to best leverage those innate qualities without even thinking about it um we're kind of reaping the same benefit to some extent that people would gain from like deep um somewhat critical or at least humble and skeptical introspection which is active inference applied to itself in a sense yeah i yeah maybe maybe it is pretty meta which i think is really cool and i just maybe maybe uh try to connect it to some things people might be familiar with i think tristan harris often speaks about like the sort of current technology which is sort of race to the bottom that it keeps like if it it sort of thrives on exploiting our different vulnerabilities cognitive vulnerabilities um and profiting off those but at this sort of the externality is that you know we're degrading human cognition here this really i feel like we're in this sort of trying to be in the space race to the top which is another kind of thing that tristan harris i've heard him yeah trying to think about technology the way that yeah it's actually making us more open-minded more attuned to sort of our space and the collective like where we're repeating like collective stance making and yeah i feel like it's really racing how do we get people how do we build technology to make people more yeah critically reflective and open-minded all those things and we have to apply those principles to do it we have to understand those things we have to know what means help bring it about yeah that's build technology open-mindedness the meta maybe a last active comment which will go to our last slide is like will what you mentioned with uh active open-mindedness which is surely a term that preceded active inference it it force ages active engagement and then the dilemma or tension or dialectic between like what we bring to the table and then what the table is providing to us and so if it were um some other kind of product design we're making a soft drink are we gonna barge in with assumptions and then what are the danger modes of that but also what is the danger of just what every single person tells you putting too much attention on the specifics overfitting to those 20 people and making some kind of like chimera that only appeals to them and that would be one thing in the food engineering and food design but this is the leverage point on digital communication and so it becomes even more immediacy because this is like the coefficient on how we communicate and share and learn nowadays and also like puts in these higher order questions like we were talking about earlier which is there's always going to be an argument for centralization and that's not to say that one extreme is better than other it's always contextual but there always is an argument to just do it faster internally rather than potentially avert a deeper failure mode but there are other challenges arising with decentralized systems so there's no generalized free lunch but then the question could be like but could there be information environments and regimes of action and regimes of that are consensual and also productive and so on so i agree it's been an awesome um and and to bring in will with yours and lauren's experience on design and seeing design as like almost the art and science and practice of bringing the counter factual and the adjacent possible via inference into something that is implemented so i i think there's a lot to connect there so maybe let's go to our last um slide and then if anybody in the live chat has any thoughts or questions otherwise um ronan looking forward to hearing these yeah just stamps yeah episodic design you feel like there's the whole like full stack episodes like there's design engineering all the and in between um but yeah so we were talking about you know some of the sort of trade-offs between the things yeah where we are now and you know centralized systems and decentralized systems there are a lot of open challenges here uh some of them we kind of already talked about so i'll go through them but yeah and just kind of finish on some of these things that we're looking at i was trying to make progress with our research proposal and development and all that so yeah i think you know one thing is definitely that we're in sort of what can we do right now to help uh clean up our information systems given that you know many of these tools don't exist yet and they're sort of i guess a lot of people would say these are like it could be a decade-long project to build these kind of interoperable protocols um so yeah where what do we do right now i think yeah if anyone has thoughts on if we're going to raise those otherwise you can just i'll run through them all if you have thoughts you can go back to it um yeah i think there's this kind of you know we talked about a few times during the conversation you have this kind of user which is the local minima of convenience so users are just like they're getting used and they're using software but but it's really like this kind of very convenient local minima and get stuck in it how do you transition towards more active sensitive role uh there's a lot of sort of culture and norms and social aspects here that we can definitely also dive deeper into but yeah probably have future conversations with that um and yeah there's no the protocol problem which we're uh mindful of where this is i think best express but it's xkcd comic you know you have some kind of uh situation where you want to introduce the new standard but they're then you already standard try to generalize everyone else's protocols and now they're like shifting protocols and you're stuck in the same situation yeah this is a problem i don't know any of you like really great results i mean i'm not aware of so many success stories here so it is clear we have like a challenge uh facing us but um on the other hand web 3 seems to be all about tackling this protocol so we feel like we're yeah in good company at least thinking about this will you want to add something or uh yeah one of the things i love the second one um and i think one of the things that resonates with me is i think it speaks to one of the original explicit goals of vo which is like obviously this is a huge problem and one of the things we were trying to do in designing that interface is to make it as convenient as possible to be an active sensemaker largely this is driven by by my own research process i have adhd i have a very hard time getting through uh research paper so i was actually like we were trying to find a way to make going through this process that seems really overwhelming to some people including myself um a lot more convenient but a lot more simple and taking a lot of that like the labor burden off so that um like the person's mind is free to think and record and doesn't have to actually manage like really what is a fairly sophisticated process for people who i mean both of you are seasoned academics you you know that it's not a trivial skill to be able to to go through loads of information and extract what's meaningful and figure out where you need to delve into new um arenas and if you can make that process that requires a lot of discipline and planning and structures um whatnot if you can make that very convenient then all of a sudden you can just think about the thing think about the information and the ideas that are in the information so of course we don't have that tool yet so what can we do right now nice thanks will just a few other um comments the interoperability we brought up earlier um the content is often highly interoperable today like whether you can download it directly or right click save or use obs to screencap pretty much any content can be delivered on another platform you can share text or an image you found on one location to another so in many ways amidst the platforms jumping from platform to platform we have interoperable content because we have like file types for content jpeg running you highlighted the interoperability of these attention markers the ones that we have today the the likes and so on but then the ones that we could have tomorrow with all these other kinds of interfaces and then i um wondered well where are we at with financial interoperability is there a way for even this one special kind of like a ledger or some kind of bookkeeping what are the interoperability mechanisms that exist for financial data and then would the knowledge case be easier or harder and then what happened this um 14 and then going to 15 so how does the financial information stay coherent with markets there isn't an interoperability layer necessarily directly between two currencies but then there are different kinds of intermediators who can be clear or obscure with why they're providing some kind of exchange and so i think that points to like information exchanges before some kind of sweeping knowledge generalization synthesis like a future knowledge protocol there would be many options for just like the apis and the kind of web scraping and re-parsing techniques exist today to to act as markets that help swap different information um and then one other point to make was talked also about like what is opaque and what is transparent and in some earlier active streams actually on like consciousness and attention and awareness we had an interesting discussion around transparency and opaque a transparent thing is something that you can't see you see through it because you can't see something that's opaque you do see because you can't see through it so what aspects of the system should be transparent which isn't to say we could investigate the source code because that would actually be opacity and one could say well it's fully documented totally comprehensive opacity well that's great but what are the parts of the system that are transparent like the water we swim in that would be not visible and then what are the parts of the system where people would see them and know them and those would be opaque yeah that's been awesome but like that's kind of mind-blowing thought um i yeah that kind of brings to mind you know little distinction making like convenience isn't always i was treating it here as sort of something that's kind of cursed but he was saying it can also be doom or benefit uh the right thing is made convenient and it kind of seems like what you're saying also is we need to make the right things like just like you know the interface like a cell phone you know the right things that are opaque and the right things are transparent where the importance is for controlling it the relevant affordances for controlling it are the ones that you see the ones you don't need to see you don't see so yeah probably something like that it seems like like again design how to make uh information landscapes that sort of show you the right levers and obscure the things you don't need to see so you can not get overwhelmed and probably being able to end this kind of openness you know the open source quality of them means that different people will want different levels so some people really want to play with the nuts and bolts and other people just want to like have like simple interfaces but allowing all that span of of interfaces i think is part of the yeah yeah it's just not like this kind of plurality aspect you were talking about i think it's really important that different people will see different interfaces based on liquids what's best for them that's one of the biggest advantages of having this interoperability is now we have an opportunity to take what is ultimately the same data set and interfaces make end user applications that are tailored to particular needs or people or use cases or accessibility um needs and whatnot so that really niche groups of people can interact on this broad scale in a way that specifically suits them that can't be accommodated with a platform or with like kind of a one-size-fits-all approach to um manipulating with engaging with interacting with this data this information information yeah one quick comment on the accessibility because i think it it really distinguishes the uh semi tectonic the building like stigma g from the attention like stigma g so content accessibility is the prerequisite for there to be the trail content accessibility could include like screen reader affordances changing the font size changing the colors um making sure that just like the open science movement has been highlighting like that the content quite literally is accessible in any form let alone one that accounts for like sensory differences but just making the content available has to be a first step what does attention accessibility look like so like if previously it was like we need that pdf accessible not behind a paywall okay content accessibility with preprint servers to a large extent in the scientific domain just speaking narrowly like that content is accessible and then what are all the trails that go from this one pre-print website and pdf and the way that these 700 people had these thoughts and these 300 people had these thoughts that's that transition amongst different content and giving visibility to that content because that content is being recorded and it is being used to shape attention but through this covert platform-owned loop rather than us being able to like be like i want to publicly add this comment on the paper and then this is just i want this to be only in my private information or something like that um but i think that really like highlights that accessibility isn't just the font size genre there's another connecting the dots kind of accessibility that doesn't exist even like on the radar because it feels like well what do you need to understand the scientific paper beyond having access to the pdf it's like you need like everything and now that can be a a clearer conversation yeah yeah i love that acceptability is broader than just the content acceptability to the reactions of people the content well any other like thoughts or directions or information you want to share feel free to go for it yeah this was great i mean just really it's fun to dive into all these things yeah so many quick questions and i just again reiterated you know it'll take a lot of minds to think about these things and work on these things a lot of different types of minds builders and all that so yeah if you're interested by any of this then we're really welcoming collaborators awesome yeah we can have like a 26.2 we can always continue the series and um like version and update and include others in the conversation so great presentation and thanks both for joining thank you for putting this on yeah thanks so much goodbye see ya okay it's done it is fun there's a few there's a few things one is to just share it and ask people but it is um it's an hour and a half long